



△ Complex-structured planetary nebula

This nebula (NGC 5189) has a complex structure, with two separate bodies of gas expanding outward in different directions. This might be explained by the presence of a second star orbiting the central white dwarf. The nebula lies about 3,000 light-years away.

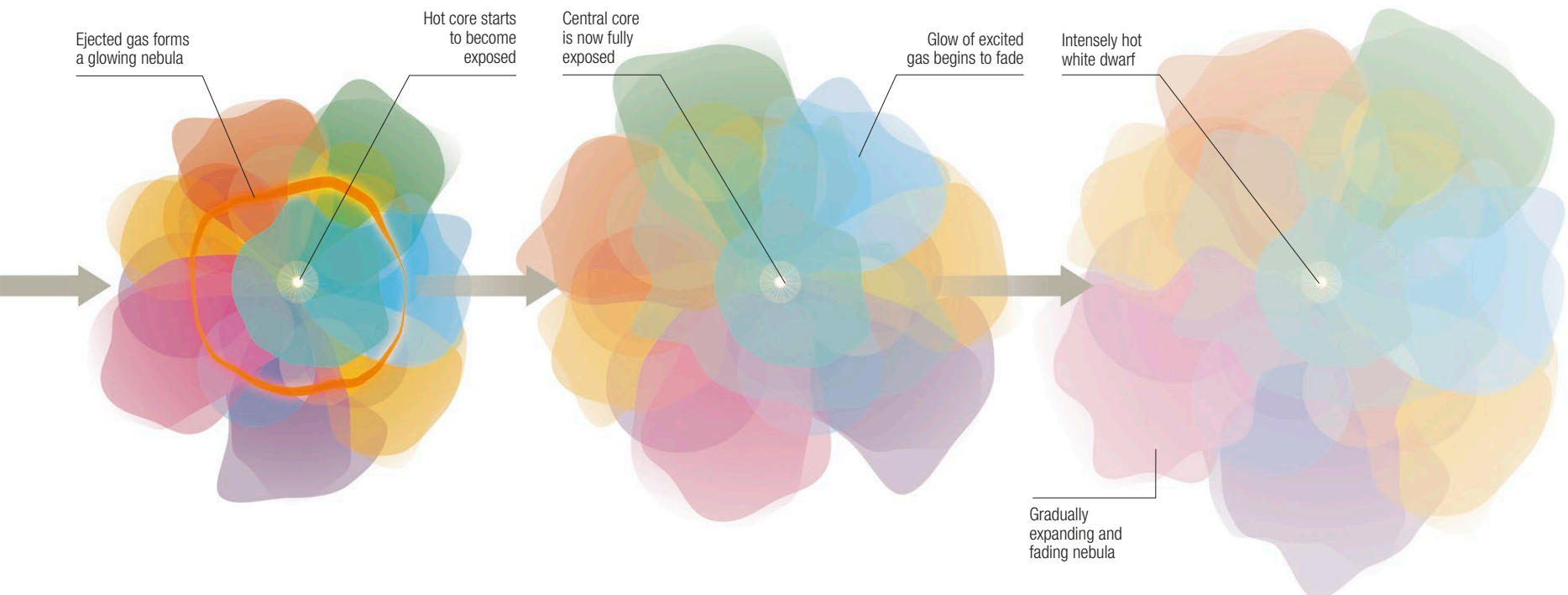
White dwarfs

When a red giant has shed all its outer layers of gas, forming a planetary nebula, what remains is a hot core consisting, in most cases, of carbon and oxygen. This object is called a white dwarf and is extremely dense—a teaspoonful of it would weigh several tons. A white dwarf also starts off extremely hot with a surface temperature of up to 270,000°F (150,000°C). However, it is not hot enough for internal nuclear fusion reactions to occur. Over extremely long periods of time, a white dwarf gradually cools and fades, eventually (it is envisaged) becoming a cold object called a black dwarf. However, the Universe is not yet old enough for any white dwarf to have cooled to the black dwarf stage.



◁ Fleming 1

This planetary nebula is highly unusual in that it contains two white dwarf stars circling close to each other at the nebula's center. Their orbital motions explain the presence of some remarkably symmetrical jets and other structures that weave into knotty, curved patterns in the surrounding gas.



4 Planetary nebula forms

As the star sheds more and more of its gas layers, its core—at this stage usually consisting largely of carbon and oxygen produced by helium fusion—becomes exposed. Intense ultraviolet radiation given off by the core heats the ejected clouds of gas, which begin to glow or fluoresce in a variety of colours due to variations in temperature.

5 Planetary nebula expands

While the nebula expands into space, the excitation from its central star begins to dwindle, and the glow from its gases starts to fade. A planetary nebula typically lasts for a few tens of thousands of years (compared to billions of years for a typical Sun-like star), and during this time it continually evolves.

6 White dwarf remains

Finally, almost all that remains is the exhausted core of the star, known as a white dwarf. Although extremely hot, it looks faint from a distance because of its small size. As the nebula's material drifts away, it becomes part of the interstellar medium—the diffuse matter that fills the space between stars in a galaxy.